

RESEARCH FOCUS

My research is motivated by firsthand experience working in data sparse regions, which led me to pursue satellite and deep learning based weather forecasting to close critical observation gaps. This focus culminated in a first place finish at the GeoNet(ICCV'23) and Weather4cast(NeurIPS'22) Challenges, and strengthened my interest in combining heterogeneous Earth observation and in situ datasets with deep learning to characterize extreme weather hazards more reliably. Through projects spanning the disaster lifecycle, from early warning to damage and impact assessment, I now focus on a unifying question: how forecasting improvements translate into measurable decision value for climate adaptation under uncertainty.

Keywords: Data-driven weather forecasting, Extreme hazard and impact profiling, and Decision value metrics for climate adaptation, but not limited to.

EDUCATION

- **Korea Advanced Institute of Science and Technology** Daejeon, South Korea
Ph.D. student of The Graduate School of Green Growth and Sustainability(advisor: Changick Kim) Sep. 2025 –
- **Ewha Womans University** Seoul, South Korea
Master of Climate and Energy System Engineering (advisor: Yong-sang Choi) Mar. 2020 – Aug. 2022
- **Ewha Womans University** Seoul, South Korea
Bachelor of Environmental Science and Engineering Mar. 2014 – Feb. 2020

PUBLICATIONS

(Equal contribution or correspondence are denoted by *)

- [1] (ICLRW) Doyi Kim, J Ahn, H McJeon, and C Kim, “Efficiency vs Demand in AI Electricity: Implications for Post-AGI Scaling,” 2026. [arxiv]
- [2] (ICLR) Doyi Kim, M Seo, and C Kim, “Station2Radar: Query-Conditioned Gaussian Splatting for Precipitation Field,” 2026. [paper]
- [3] (AAAI Oral) Y Park, M Seo, Doyi Kim, and Y Choi, “Data-driven Precipitation Nowcasting Using Satellite Imagery,” Proceedings of the AAAI Conference on Artificial Intelligence. 2025. [paper]
- [4] Doyi Kim, and Y Choi, “Utilization of satellite imagery and artificial intelligence for disaster management: approaches and case studies,” ITU Journal on Future and Evolving Technologies. 2025.[paper]
- [5] Doyi Kim, M Seo, and Y Choi, “ACE Metric: Advection and Convection Evaluation for Accurate Weather Forecasting,” 2024. [arXiv]
- [6] (ICLR Spotlight) Y Park, M Seo, Doyi Kim, H Kim, S Choi, B Choi, J Ryu, S Son, H Jeon, and Y Choi, “Long-Term Typhoon Trajectory Prediction: A Physics-Conditioned Approach Without Reanalysis Data,” In the 12th International Conference on Learning Representations (ICLR). 2024. [paper]
- [7] (ECCV) D Yoon, M Seo, Doyi Kim, Y Choi, and D Cho, “Probabilistic Weather Forecasting with Deterministic Guidance-based Diffusion Model,” In the 18th European Conference on Computer Vision (ECCV). 2024. [paper]
- [8] Gruca, A., Serva, F., Lliso, L., Rípodas, P., Calbet, X., Herruzo, P., ..., Doyi Kim, ..., and Kreil, D. P., “Weather4cast at neurips 2022: Super-resolution rain movie prediction under spatio-temporal shifts,” In NeurIPS 2022 Competition Track (pp. 292-313). PMLR. 2023. [paper]
- [9] (SCI(E) Paper) Doyi Kim, H Kim, and Y Choi, “Unsupervised Clustering of Geostationary Satellite Cloud Properties for Estimating Precipitation Probabilities of Tropical Convective Clouds,” Journal of Applied Meteorology and Climatology 62(8), (pp. 1083-1094). 2023. [paper]
- [10] (NeurIPS Workshop) M Seo, Y Oh, Doyi Kim, D Kang, and Y Choi, “Improved Flood Insights: Diffusion-Based SAR to EO Image Translation,” NeurIPS 2023 Workshop: Tackling Climate Change with Machine Learning. [poster]
- [11] (NeurIPS Workshop) Y Oh, M Seo, Doyi Kim, and J Seo, “Prototype-oriented Unsupervised Change Detection for Disaster Management,” NeurIPS 2023 Workshop: Tackling Climate Change with Machine Learning. [poster]
- [12] Doyi Kim*, M Seo*, J Seo, and H Lee, “Masked Autoregressive Model for Weather Forecasting,” 2023. [arXiv]
- [13] Doyi Kim, Y Choi, E Cho, S Shin, J Ryu, and H Jeong, “Forecasting and Tracking Convective Clouds based on the Deep Learning Method using Geostationary Satellite Imageries,” 103rd AMS Annual Meeting, 2023.
- [14] Doyi Kim, Y Choi, M Seo, S Shin, and H Jeong, “Short-term forecasting of typhoon rainfall with a deep-learning-based disaster monitoring model,” Environmental Data Science, 2023. [paper]

- [15] M Seo, Doyi Kim, S Shin, E Kim, S Ahn, and Y Choi, "Simple Baseline for Weather Forecasting Using Spatiotemporal Context Aggregation Network," 2022. [arXiv]
- [16] M Seo, Doyi Kim, S Shin, E Kim, S Ahn, and Y Choi, "Domain Generalization Strategy to Train Classifiers Robust to Spatial-Temporal Shift," 2022. [arXiv]

EXPERIENCE

- **SI Analytics** Daejeon, South Korea
Earth Intelligence Division, AI Research Center, Research Scientist Jul 2022 - Aug 2025
 - **Satellite-to-Radar Image Generation:** Developed deep learning models to transform satellite imagery into radar-like precipitation data, enhancing real-time precipitation estimates.
 - **Global Precipitation Forecasting:** Led global precipitation estimation and forecasting efforts using multi-geostationary satellite imagery and calibration techniques.
 - **Disaster Detection and Impact Assessment:** Implemented change detection models to identify affected areas post-disaster through optical satellite imagery and deep learning.
 - **International Corporation in Weather and Disaster Services:** Strategically planned applications of multi-satellite imagery and AI for national and international projects focused on weather prediction and disaster damage detection.
 - **Data Generalization and Augmentation:** Proposed domain generalization methods to bridge observational data gaps across regions and developed data augmentation strategies for meteorological datasets.
- **Secretariat of the Pacific Regional Environment Programme (SPREP, UNEP)** Apia, Samoa
Climate Change Resilience-Pacific Meteorology team, Intern Mar 2020 - Jul 2020
 - **Early Warning System Support:** Support for the establishment of early warning systems in Pacific small islands and preparation of fact sheet for sector-specific stakeholders
 - **Community-based Warning Solution:** Design of media utilization strategies to improve the delivery of disaster warnings to last-mile populations

PROJECTS

- **CIEWS Project Development with AI-based Satellite Image Analysis Technologies** Nov 2023 - Nov 2024
Korea Environmental Industry and Technology Institute
 - Led design and planning for international climate adaptation projects funded by the Green Climate Fund (GCF)
 - Key projects: Climate Information Services for Resilient Development Planning in Vanuatu, Nepal Satellite-Based Landslide Early Warning and Damage Detection System, Multi-Hazard Early Warning System in Pacific Island Countries
 - Responsibilities: Stakeholder engagement, early warning system design, and funding proposal development; successfully advanced projects from concept to funding proposal stage
- **Satellite-based Convective Clouds Diagnostics and Prediction** Jul 2022 - Dec 2023
National Meteorological Satellite Center
 - Developed and implemented a predictive model (CrevNet) to detect and track convective cloud cells through geostationary satellite imagery.
 - Applied video frame prediction with multi-GPU implementation to improve forecasting of weather events, enhancing computational efficiency.
 - Responsibilities: dataset preprocessing, code optimization, and experimental results analysis
- **Development of Climate and Environment Risk Management Tool(Frontier-1.5D)** Oct 2020 - May 2022
Financial Supervisory Service of Korea
 - Led the development of a climate stress test model and scenarios to forecast capital loss in enterprises due to policy shifts (climate change regulations, greenhouse gas reduction mandates).
 - Responsibilities: Disaster loss data collection, disaster-loss relationship function development, and Excel-based risk assessment tool creation

AWARDS & HONORS

- **Named in the "Geospatial World Rising Stars 50"**, Geospatial World, 2024.
- **Awarded the AI/ML Solutions for Climate Change Innovation Factory Challenger**, ITU & UNESCO, 2023.
- **Winner of GeoNet Challenge on Geographical Robustness**, ICCV, 2023.
- **Winner of Weather4Cast: Super-resolution rain movie prediction under spatio-temporal shifts**, NeurIPS, 2022.
- **Awarded the Future Leaders Scholarship**, Hyundai Motor Chung Mong-Koo Foundation, 2021-2022.
- **Excellence award of Zero Energy Building Business model Competition**, Korean Energy Agency, 2018.

PATENTS

- **Method of image-to-image translation using diffusion model**
United States. 18/769.297 (Application) 2024
- **Method for combining probabilistic and deterministic models for prediction**
Republic of Korea. 10-2024-0071255 (Application) 2024
- **Method for predicting typhoon paths using time series weather data prediction models**
Republic of Korea. 10-2024-0071256 (Application) 2024
- **Method for detecting changes for disaster management**
Republic of Korea. 10-2024-0056850 (Application) 2024
- **Method for analyzing satellite image using spatiotemporal context information**
Republic of Korea. 10-2023-0094175 (Application) 2023
- **Method for predicting frame using deep learning**
Republic of Korea. 10-2023-0115699 (Application) 2023

EXTRA

- **She Space International Program(Space4Women) Mentor**
Korea Aerospace Research Institute 2023-2025
- **Certified International Meteorological Expert Training**
Korea Meteorological Administration 2019
- **Global Student Forum 13th Program Manager**
World Engineering Education Forum (WEEF) 2017
- **Young Engineers Honor Society Vice-President**
The National Academy of Engineering of Korea 2016-2019